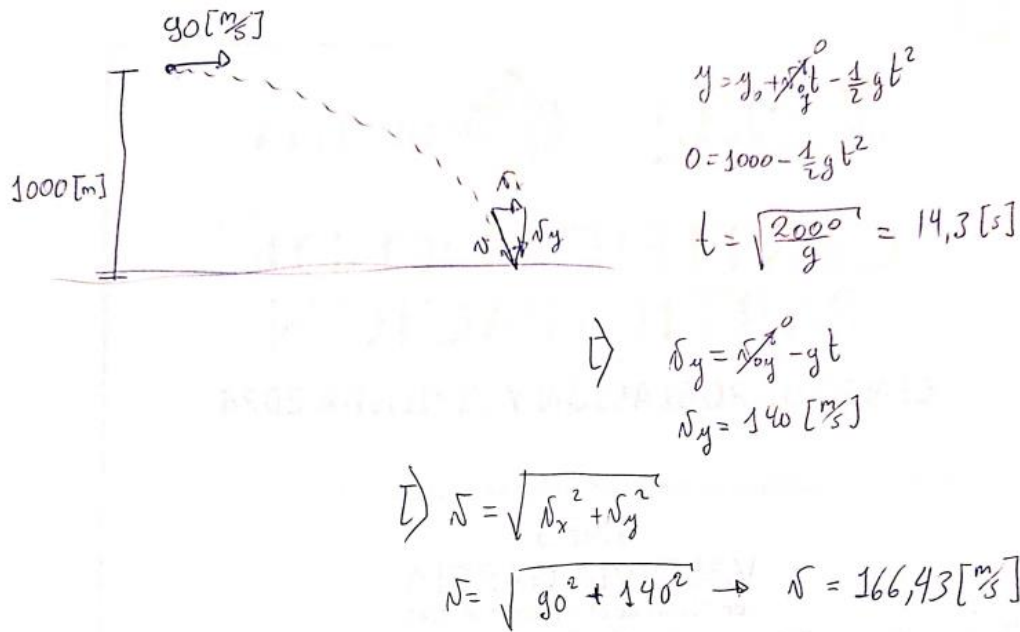
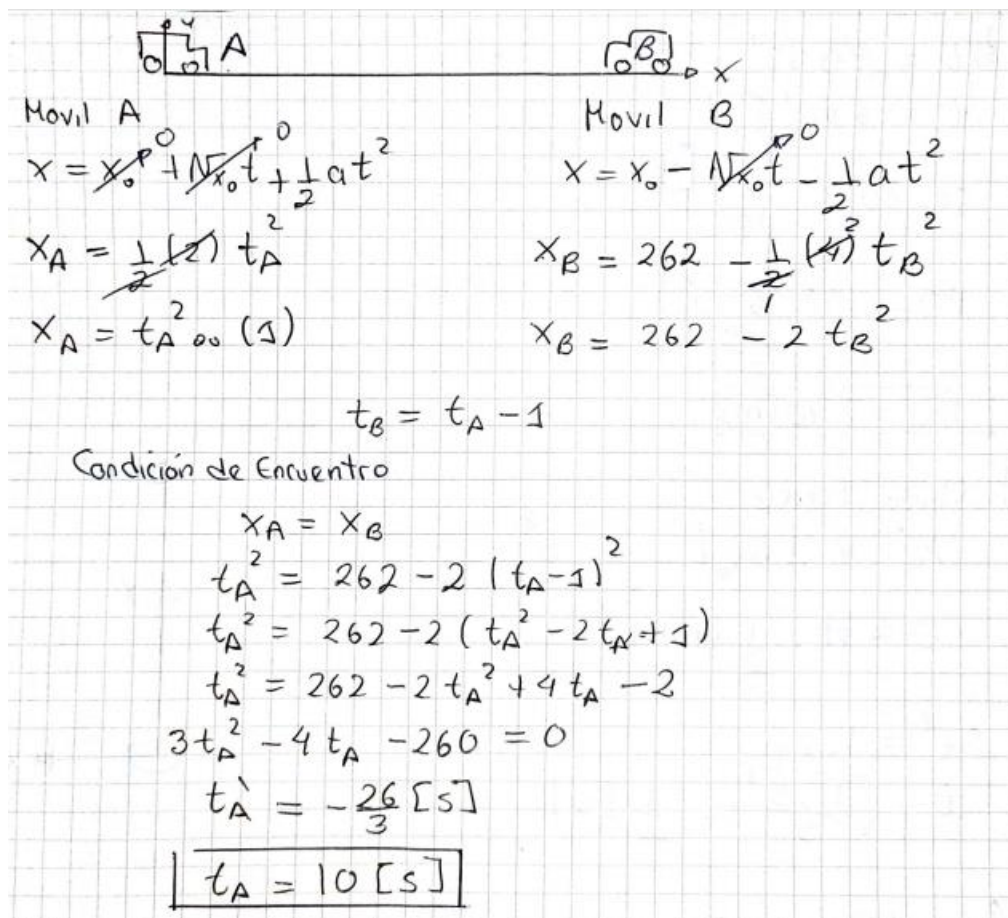


F1



F2



F3

c) Queda libre

$$y = y_0 - v_{y0} t - \frac{1}{2} g t^2$$

$$0 = H - \frac{1}{2} (10) t^2$$

$$5 t^2 = 1.25$$

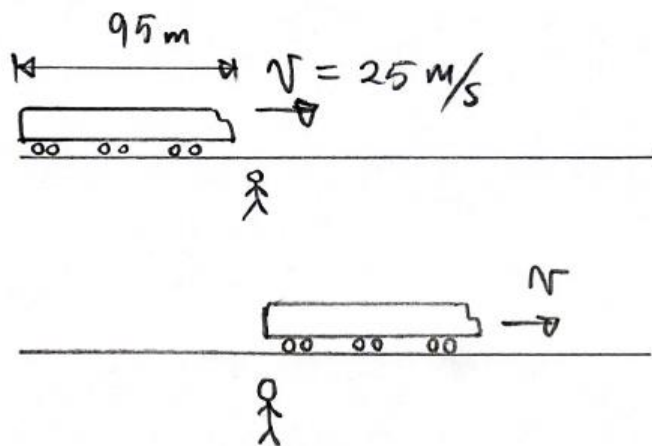
$$t = \sqrt{\frac{1.25}{5}} = \frac{1}{2} [s]$$

$$\omega = \frac{\Delta \theta}{\Delta t} = \frac{2\pi}{\frac{1}{2}}$$

$$\omega = 4\pi \left[\frac{\text{rad}}{s} \right]$$



F4



$$v^2 = 2a(180)$$

$$25^2 = 2a(180)$$

$$a = \frac{25^2}{2(180)}$$

$$a = \frac{625 \cdot 125}{360}$$

$$a = \frac{125}{72} \text{ m/s}^2$$

$$v^2 = 25^2 + 2a(95)$$

$$v^2 = 625 + 2 \cdot \frac{125}{72} \cdot 95$$

$$v = 30,9 \text{ m/s}$$

F5

$$R_c = 20 \text{ cm} \quad 80 \text{ pedaleadas} \times \frac{1 \text{ vuel}}{2 \text{ pedale}} \\ R_p = 8 \text{ cm} \quad 40 \text{ vuel/min}$$

$$R_R = 60 \text{ cm}$$

$$\omega_c = 40 \text{ vuel/min} \times \frac{2\pi}{1 \text{ vuel}} \times \frac{1 \text{ min}}{60 \text{ s}}$$

$$\omega_c = 4,19 \text{ rad/s} = \frac{4}{3} \pi$$

$$v_c = R_c \omega_c \rightarrow \frac{4}{3} \pi$$

$$v_c = 20(4,19)$$

$$v_p = v_c$$

$$v_p = \omega_p R_p$$

$$\omega_p = \frac{v_p}{R_p} = \frac{20(4,19)}{8} = \frac{10\pi}{3}$$

$$\omega_p = 10,47 \text{ rad/s}$$

$$\omega_p = \omega_R$$

$$v_R = R_R \omega_R \quad \rightarrow \frac{6}{10} \times \frac{10}{3} \pi$$

$$v_R = 0,6(10,47)$$

$$v_R = 2\pi = 6,28 \text{ m/s}$$